

Literature Status for arXiv:math/0406477: Exact Complexity

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Status

This is a `literature_already_answered` packet. It records that the exact-complexity question raised in Remark 5.4 of Ferenczi–Medina Galego, *Some equivalence relations which are Borel reducible to isomorphism between separable Banach spaces*, arXiv:math/0406477, was answered by Ferenczi–Louveau–Rosendal, *The complexity of classifying separable Banach spaces up to isomorphism*, arXiv:math/0610289.

Source Question

Remark 5.4 of arXiv:math/0406477 says that the exact complexity of isomorphism between separable Banach spaces, viewed as subspaces of $C[0, 1]$ with the Effros–Borel structure, was “quite open”. It also says the same was true “even” for Banach spaces with Schauder bases, viewed as subsequences of the universal basis of Pelczynski, and suggests the possibility that the relation could be Σ_1^1 -complete, i.e. maximum among analytic equivalence relations under Borel reducibility.

Supporting Answer

The abstract of arXiv:math/0610289 states that isomorphism between separable Banach spaces is a complete analytic equivalence relation, meaning that every analytic equivalence relation Borel reduces to it. The main theorem in the section on isomorphism of separable Banach spaces repeats this statement:

isomorphism between separable Banach spaces is complete analytic.

This is exactly the Σ_1^1 -complete alternative raised in the source remark.

The supporting paper also addresses the basis-side concern from the same remark. Theorem 1 proves that permutative equivalence between, even unconditional, Schauder bases is a complete analytic equivalence relation. Moreover, the later Banach-space reduction uses the spaces $\Delta(Z_S, 2)$, which are reflexive and carry suppression unconditional bases. Thus the later theorem answers the general separable Banach-space exact-complexity question and supplies the expected basis-level complete analytic phenomenon.

Scope

This packet does not address the separate questions in Remark 5.2 and Conjecture 5.3 of arXiv:math/0406477: ergodicity of ℓ_p for $p > 2$, the conjecture that ℓ_2 is the only non-ergodic Banach space, and

the block-subspace dichotomy for Banach spaces with unconditional bases remain outside this literature-status claim.

Search Evidence

The lane-12 queue selected arXiv:math/0406477. Cheap run indexes were searched for the arXiv id, title words, Ferenczi–Galego, Borel reducibility, and exact-complexity phrases; no duplicate packet for this source was found. Targeted web and local searches for phrases including **isomorphism between separable Banach spaces complete analytic**, **Σ^1_1 -complete**, **Schauder bases**, and **universal basis of Pelczynski** identified arXiv:math/0610289 as the decisive later paper.

References

- [1] V. Ferenczi and E. Medina Galego, *Some equivalence relations which are Borel reducible to isomorphism between separable Banach spaces*, arXiv:math/0406477.
- [2] V. Ferenczi, A. Louveau, and C. Rosendal, *The complexity of classifying separable Banach spaces up to isomorphism*, arXiv:math/0610289.